In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('custdata.tsv', sep='\t')
cols = ['age', 'income', 'num.vehicles']
x = df[cols]
print(df.shape)
x.describe().round(2)
```

```
(1000, 11)
```

Out[1]:

	age	income	num.vehicles
count	1000.00	1000.00	944.00
mean	51.70	53504.77	1.92
std	18.86	65478.07	1.10
min	0.00	-8700.00	0.00
25%	38.00	14600.00	1.00
50%	50.00	35000.00	2.00
75%	64.00	67000.00	2.00
max	146.68	615000.00	6.00

Age ranges from 0 to ~147, income has a negative minimum, and num.vehicles has missing values (count < 1000).

In [2]:
```
before = x.corr()
before.round(3)
```

Out[2]:

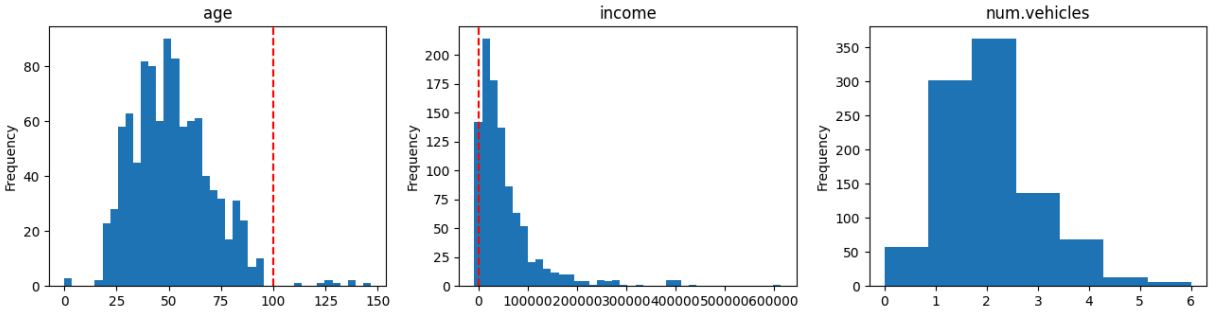
	age	income	num.vehicles
age	1.000	0.027	-0.064
income	0.027	1.000	0.139
num.vehicles	-0.064	0.139	1.000

In [3]:
```
print('age == 0          :', (x.age == 0).sum())
print('age > 105         :', (x.age > 105).sum(), '-> values', sorted(x.age[x
print('income < 0        :', (x.income < 0).sum(), '-> values', x.income[x.ir
print('income == 0       :', (x.income == 0).sum(),)
print('num.vehicles NaN :', x['num.vehicles'].isna().sum())
print('oldest plausible age:', x.age[x.age <= 105].max().round(1))

fig, ax = plt.subplots(1, 3, figsize=(13, 3.5))
x.age.plot.hist(bins=40, ax=ax[0], title='age'); ax[0].axvline(100, color='r
x.income.plot.hist(bins=40, ax=ax[1], title='income'); ax[1].axvline(0, colo
```

```
x['num.vehicles'].plot.hist(bins=7, ax=ax[2], title='num.vehicles')
plt.tight_layout(); plt.show()
```

```
age == 0          : 3
age > 105         : 8 -> values [113.3, 123.1, 124.8, 126.7, 130.4, 136.1, 13
7.7, 146.7]
income < 0        : 1 -> values [-8700]
income == 0       : 78
num.vehicles NaN : 56
oldest plausible age: 93.0
```



In [4]:
```
valid = (x.age > 0) & (x.age <= 105) & (x.income >= 0)
clean = x[valid].dropna()

removed = pd.Series({
    'age == 0': (x.age == 0).sum(),
    'age > 105': (x.age > 105).sum(),
    'income < 0': (x.income < 0).sum(),
    'missing num.vehicles': x['num.vehicles'].isna().sum(),
})
print(removed.to_string())
print(f'\nRows: {len(x)} -> {len(clean)}  ({len(x) - len(clean)} removed)')
```

```
age == 0                 3
age > 105                8
income < 0               1
missing num.vehicles    56

Rows: 1000 -> 932  (68 removed)
```

What was removed and why

I removed age = 0(3, a newborn can't be a customer and 0 is likely a placeholder), age > 105 (8, too high, many of the ages are impossible and anything over 105 is very improbable), negative income(1, not valid. You can't have a negative income), and missing `num.vehicles` (56, filling pulls correlations toward zero and understates relationships). 68 removed total.

In [5]:
```
after = clean.corr()
after.round(3)
```

Out[5]:

	age	income	num.vehicles
age	1.000	-0.003	-0.066
income	-0.003	1.000	0.138
num.vehicles	-0.066	0.138	1.000

In [6]:
```
diff = (after - before).round(3)
pairs = [('age','income'), ('age','num.vehicles'), ('income','num.vehicles')
pd.DataFrame({
    'before': [before.loc[a, b] for a, b in pairs],
    'after':  [after.loc[a, b] for a, b in pairs],
    'change': [diff.loc[a, b] for a, b in pairs],
}, index=[f'{a} ~ {b}' for a, b in pairs]).round(3)
```

Out[6]:

	before	after	change
age ~ income	0.027	-0.003	-0.030
age ~ num.vehicles	-0.064	-0.066	-0.003
income ~ num.vehicles	0.139	0.138	-0.001

Sensitivity: also drop income = 0

In [7]:
```
alt = x[(x.age > 0) & (x.age <= 105) & (x.income > 0)].dropna()
print('n =', len(alt))
alt.corr().round(3)
```

```
n = 879
```

Out[7]:

	age	income	num.vehicles
age	1.000	-0.023	-0.072
income	-0.023	1.000	0.148
num.vehicles	-0.072	0.148	1.000

Interpretation

Age and income are uncorrelated(r ≈ 0.00). Income has a weak positive relationship with number of vehicles (r ≈ 0.14) and age has a very weak negative relationship with vehicles (r ≈ -0.07). Neither is strong enough to be useful for prediction.